



Selected Topics in Cryptography

Lab 05: Toy ECDSA

March 17, 2026

Develop the following exercises by your own. You can use C/C++, C#, Java or Python to develop your source code, but please use only one of this languages to develop all the exercises. You **MUST NOT USE any cryptographic library to implement ECDSA**. Use the functions you implemented before to do arithmetic on the rational points of a non-singular elliptic curve.

1 Key generation for ECDSA

1. Design and implement a function that receives as parameters:

- p a prime number
- a, b coefficients of a non-singular elliptic curve
- G a generator in the chosen elliptic curve of order prime q

The output of this function must be a pair of keys for ECDSA: the public key $k_{pub} = (p, a, b, q, G, B)$ and the private key $K_{priv} = d$, where $0 < d < q$ must be randomly chosen.

2. Use the previous function to generate your own pair of keys for ECDSA. Use the information in the given file: **EC_primeorder**
3. Upload the parameters of your public key to the Excel file (see link)

2 Signature generation

1. Design and implement a function to generate the signature for an integer $1 \leq m \leq q - 1$. Your function must receive as a parameter the integer $0 < m < q$ and the public parameters of the public key q, G and the private key d . Then do the following
 - (a) Choose a secret random key $0 < K_E < q$.
 - (b) Compute $T = K_E G \bmod q$
 - (c) Take $r = x_R$
 - (d) Compute $s = (m + dr)K_E^{-1} \bmod q$
 - (e) Print the signature (r, s)
2. Choose an integer $0 < m < q$ and generate the signature (r, s) for m . Upload the message m and the signature (r, s) to the Excel file.

3 Verification signature

1. Design and implement a function to verify a signature. Your function must receive as parameters: the public key $k_{pub} = (p, a, b, q, G, B)$, the message m and the signature (r, s) . Then do the following:
 - (a) Compute $w = s^{-1} \bmod q$
 - (b) Compute $u_1 = wm \bmod q$
 - (c) Compute $u_2 = wr \bmod q$
 - (d) Compute $P = u_1G + u_2B \bmod q$
 - (e) If $x_P == r \bmod q$ the signature is valid, otherwise it is not.
2. Use the data listed in the Excel file: m and (r, s) and the previous function to verify signatures in the Excel file

4 Questions

Answer the following questions

1. Use the mathematical expressions of ECDSA involved in key generation and signature generation to show that if the signature is valid, then $v == r \bmod q$. Write down all the details.
2. What would happen if q is not a prime number?
3. Solve the discrete logarithm problem to find the private keys, using the information of the public keys in the Excel file.

5 Products

Every student must write her/his own report including the following information:

1. Personal information, date of the lab session and the topic that we are studying in this lab session.
2. Source code to generate a key pair for ECDSA
3. Your results to verify the signatures in the Excel file, i.e a list including m , (r, s) and if the signature is valid or not.